Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

A Report on

"INTERNSHIP"

Submitted in partial fulfilment for the Award of Diploma in

COMPUTER SCIENCE AND ENGINEERING

by

Manjunath G P

(102CS23029)

Under the Supervision of

Mr. Namith H S

Scontinent Technologies Pvt Ltd, Bengaluru

Under the Guidance of

Mr. Raghavendra

Cohort Owner, CPS&S Dept

SJ Govt Polytechnic, Bengaluru

Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

A Report on

"INTERNSHIP"

Submitted in partial fulfilment for the Award of Diploma in

COMPUTER SCIENCE AND ENGINEERING

by

Manjunath G P

(102CS23029)

Under the Supervision of

Mr. Namith H S

Scontinent Technologies Pvt Ltd, Bengaluru

Under the Guidance of

Mr. Raghavendra

Cohort Owner, CPS&S Dept

SJ Govt Polytechnic, Bengaluru

Government of Karnataka

GOVERNMENT POLYTECHNIC
DEPARTMENT OF TECHNICAL EDUCATION KARNATAKA

Department of Collegiate & Technical Education

SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC

BENGALURU - 560001

2025-26

CERTIFICATE

This is to certify that **Report** submitted by **Manjunath G P**, (**102CS23029**) represents his work during the **2025-26** academic year at **Scontinent Technologies Pvt Ltd, Bengaluru**, under the Specialization Pathway **"INTERNSHIP(AI&ML)-20CS61S"** fulfilling the requirements for the award of the Diploma in Computer Science and Engineering.

Cohort Owner	Program Coordinator	Principal
Mr Raghavendra M Y	**Mrs. Haritha S M**	**Sri Harish**

Examiners:

1._____ 2. _____

Scontinent Technologies Pvt Ltd.
NO. 23, Mezzanine Floor, Outer Ring Rd, 2nd Block, 2nd Stage, Naagarabhaavi, Bengaluru, Karnataka 560072
office@scontinent.com
www.scontinent.com
+91 9353548879

Internship Completion Certificate

This is to certify that **Manjunath G P**, holding the Registration Number **102CS23029**, from **Sri Jaya chamarajendra Government Polytechnic** has successfully completed their internship with **Scontinent Technologies Private Limited.**

During their tenure from **22/DEC/2025 to 10/APR/2026**, **Manjunath G P actively** participated in the **AI/ML**, demonstrating exceptional dedication, technical acumen, and strong willingness to learn. Their performance consistently exceeded expectations, reflecting their ability to grasp complex concepts and apply them effectively in practical scenarios.

Manjunath G P exhibited a high degree of professionalism, adhering to organizational processes and maintaining excellent teamwork and communication skills. They consistently met project deadlines with precision and delivered quality results, contributing to the overall success of the assignments undertaken.

We extend our best wishes to **Manjunath G P for** continued success and growth in their career. Should you require any further information, please feel free to contact us at **office@scontinent.com** or **9353548879.**

Chief Marketing Officer,
Naveen Kumar
Scontinent Technologies Pvt Ltd
Date-10-04-2026

Ministry of MSME, Govt. of India
#startupindia

ACKNOWLEDGEMENT

I extend my deepest gratitude to Sri Jayachamrajendra Government Polytechnic College, Department of Computer Science and Engineering, for granting me the invaluable opportunity to complete this internship as part of my 6th Semester Diploma curriculum.

I am profoundly thankful to Sri Harish M, Principal, for his unwavering encouragement and for fostering a conducive academic environment.

I express my sincere appreciation to Mrs. Haritha S M, Head of the Department, for her insightful guidance and steadfast support throughout this internship.

I am particularly indebted to Mr. Raghavendra M Y, Cohort Owner, for his expert guidance, motivational support, and constructive feedback that were instrumental in the successful execution of this internship.

I also wish to acknowledge Mr. Kishore Kumar, Placement Officer, for his pivotal role in securing this internship and facilitating essential industry exposure.

My heartfelt thanks go to Scontinent Technologies Pvt Ltd and its entire team for graciously providing me with this prestigious internship opportunity. I am especially grateful to Mr. Vinay K H CEO of the Organization, for his visionary leadership and profound insights. I further extend my sincere thanks to Namith H S Team Lead, for his dedicated technical guidance and unwavering support during my tenure.

I would like to extend my heartfelt thanks to my Parents and Well-Wishers for their constant encouragement and unwavering support. Their love, belief, and sacrifices have been the driving force behind my success in this internship.

I am immensely grateful to my friends, whose constant support and collaborative spirit have made this journey truly enjoyable.

Manjunath GP

102CS23029

EXECUTIVE SUMMARY

This internship at Scontinent Technologies Pvt Ltd in Bengaluru was centered on the practical application of Artificial Intelligence and Machine Learning technologies to address real-world challenges in the financial services and human resource management domains. This report presents a thorough account of the 16-week On Job Training experience, detailing the organizational profile, skill development activities, technical project implementations, and key professional insights gained during the internship tenure.

The internship was structured around two significant project deliverables. The first project involved the development of an Financial AI System for automated credit risk assessment, which employed ensemble learning techniques. The second project focused on building an AI Resume Parser that leveraged Named Entity Recognition and skill matching algorithms to transform unstructured resume documents into structured candidate profiles with quantitative compatibility scores.

Additionally, the training period fostered a comprehensive understanding of the diverse business challenges and requirements encountered by technology firms serving multiple industry verticals. Collaborative development sprints offered practical insight into customizing AI-powered solutions to address specific client needs and generate measurable business value.

In summary, this internship at Scontinent Technologies provided a transformative learning experience at the intersection of machine learning, natural language processing, and full-stack software engineering. The technical proficiency and professional competencies cultivated during this period establish a robust foundation for advancing a career in the artificial intelligence industry

TABLE OF CONTENTS

CHAPTER - 1: COMPANY OVERVIEW

Section No	Section Name	Page No
1.1	Overview of the Organization	1
1.2	Vision and Mission of the Organization	2
1.3	Growth Initiatives	3
1.4	Organization Structure	4
1.5	Roles and Responsibilities of personnel in organization	4
1.6	Products & Market Performance	5

CHAPTER - 2: ON JOB TRAINING- 1

2. Task: FINANCIAL AI

Section No	Section Name	Page No
2.1	Role	7
2.2	Job Overview	7
2.3	Future scope of the Project	8
2.4	Goal of the Project	9
2.5	Problem that Software will solve	10
2.6	Workflow of the Model	11
2.7	Tasks Assigned 5	12
2.8	The Main Algorithm used in the project	13
2.9	Project Outcomes	13

CHAPTER - 3: ON JOB TRAINING- 2

3. TASK: AI Resume Parser

Section No	Section Name	Page No
3.1	Role	15
3.2	Job Overview	16
3.3	Future scope of the Project	16
3.4	Goal of the Project	16
3.5	Problem that Software will solve	17
3.6	Workflow of the Model	17
3.7	Algorithms used in project	19
3.8	Outcomes	19

CHAPTER- 4

Section No	Section Name	Page No
4.1	Use Case – 1: Explainable Financial AI System	20
4.2	Use Case – 2: AI Resume Parser	27

Section No	Section Name	Page No
5	Resume	34
6	Photo Gallery	35
7	References	36
8	Appendices	37

LIST OF FIGURES

Figure No.	Figure Name	Page no
1.1	Company Logo	1
1.2	Organisation Structure	4
1.3	SAM AI Digital Learning Platform Logo	6
2.1	Finance AI Workflow Model	11
3.1	AI Resume Parser Workflow Model	17
4.1	Financial AI Interface	23
4.2	Financial AI Prediction	24
4.3	Financial Ai Model Evaluation Scores and Graphs	25
4.4	AI resume Parser Interface	30
4.5	AI resume Parser Result Interface	31
4.6	AI Resume Parser Model Evaluation Scores and Graphs	32
6.1	Group Photo of Interns at Scontinent Technologies Company	35
6.2	Snapshot of Interns working at Scontinent technologies	35
6.3	Snapshot of Interns At Conference Room at Scontinent Technologies	35

ABBREVATIONS

- **CEO**: Chief Executive Officer
- **CTO**: Chief Technology Officer
- **COO**: Chief Operations Officer
- **CFO**: Chief Financial Officer
- **CMO**: Chief Marketing Officer
- **MD**: Managing Director
- **HR**: Human Resources
- **MOU**: Memorandum of Understanding
- **CIE**: Continuous Internal Evaluation
- **AI**: Artificial Intelligence
- **ML**: Machine Learning
- **NLP**: Natural Language Processing
- **NER**: Named Entity Recognition
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **ROC-AUC**: Receiver Operating Characteristic
- **SVM**: Support Vector Machines
- **XGBoost / LightGBM**: Advanced Gradient Boosting
- **API**: Application Programming Interface
- **SaaS**: Software as a Service
- **CI/CD**: Continuous Integration / Continuous Deployment

- **HTML / HTML5**: HyperText Markup Language
- **CSS / CSS3**: Cascading Style Sheets
- **JS**: JavaScript
- **JSON**: JavaScript Object Notation
- **SQL / MySQL / SQLite**: Structured Query Language
- **IDE / VS Code**: Integrated Development Environment

  Visual Studio Code
- **IT**: Information Technology
- **EdTech**: Educational Technology
- **PDF**: Portable Document Format
- **DOCX**: Microsoft Word Open XML Format
- **CAD**: Computer-Aided Design

CHAPTER – 1

COMPANY PROFILE

This chapter includes an overview of Scontinent Technologies Pvt. Ltd., its vision and mission, and organizational structure. It also describes the roles and responsibilities of personnel, the company's growth initiatives, and its major products and services such as the SAM AI Digital Learning Platform and enterprise solutions.

1.1 - Overview of the Organization

Fig 1.1 Company Logo

Mr. Vinay K H, CEO is the Founder of Scontinent Technologies Pvt Ltd Scontinent Technologies Pvt Ltd was Founded in 2019 and formally incorporated as a private limited company on August 5, 2021,

Scontinent Technologies Pvt Ltd operates in the Information Technology and Educational Technology industry sector, which is one of the most dynamic and rapidly evolving sectors in the Indian and global economy.

Current Address: 32, Santosh Soudha, Near Bank of India ATM, Vidyanagar Cross, Bettahalasur, Bangalore, North Bangalore - 562157.

Contact: 9164563662,

Email: office@studentcontinent.com,

Website: https://www.scontinent.com

1.1.1 Scontinent Technologies Are:

The organization's comprehensive portfolio spans multiple vertically integrated technology domains:

- Digital Learning Platforms and Educational Technology Solutions
- Enterprise Web Application Development and Deployment
- Native Android Mobile Application Engineering
- Artificial Intelligence and Machine Learning Consulting
- Cloud-based Infrastructure and SaaS Products

Scontinent Technologies operates in the field of Computer and related activities with an authorized share capital of Rs. 10,00,000 and paid-up capital of Rs. 8,50,000 as of March 31, 2022, indicating reasonable financial strength for its operations.

1.2 - Vision and Mission of the Organization

Vision Statement: To be the global benchmark for multidisciplinary digital ecosystems, where technology helps to empower the next generation of leaders with the skills, connections, and innovation required to solve the complex challenges of the 21st century..

Mission Statement: To bridge the gap between academia and industry by providing a collaborative digital product that empowers, connects, and grows through innovative technology and multidisciplinary knowledge exchange.

Teamwork and Collaboration are fundamental to the company's operational philosophy. Cross-functional teams work together on projects, sharing knowledge and expertise to achieve common goals

1.3 Growth Initiatives

1.3.1. Strategic Partnerships:

Scontinent Technologies has aggressively pursued strategic partnerships as a cornerstone of its growth strategy. By signing MOUs with over twenty educational institutions across Karnataka and neighboring states, the company has established a reliable distribution channel for its digital learning platform..

1.3.2. Startup Ecosystem Integration:

The company has actively collaborated with more than thirty startup ventures, providing technology solutions that range from minimum viable product development to full-scale enterprise application engineering. These collaborations expose Scontinent Technologies to diverse technology stacks, business models, and market requirements, enriching the company's technical capabilities and broadening its industry expertise.

1.3.3. Recognition and Awards:

The organization's growth trajectory has been recognized through its selection for the Elevate Grand Finale 4th Round under the Karnataka Startup Cell. This competitive selection among hundreds of startups across the state validated the company's business model, technology approach, and market potential.

1.4 Organization Structure

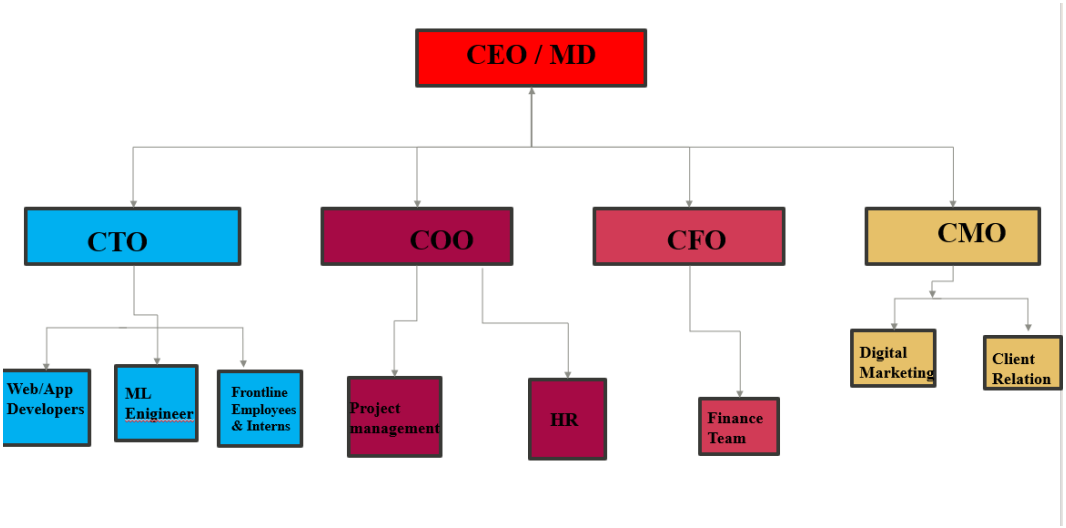

Fig 1.2. Organization Structure

Scontinent Technologies operates with a focused leadership team comprising directors who oversee the strategic and operational directions of the company. The organizational structure is designed to promote innovation, transparency, and rapid decision-making through a relatively flat hierarchy that enables direct communication between different functional levels.

1.5 Roles & Responsibilities of Personnel in Organization

At the executive level, the company is led by Mr. Kataveeranahalli Hanumantharayappa Vinay, who serves as the **Chief Executive Officer**. He is responsible for overall strategic direction, business development, and stakeholder management.

The Chief Technology Officer (CTO) manages the technology roadmap and AI/ML strategy for the company.

The Chief Operations Officer (COO) handles day-to-day operations and ensures smooth project delivery across departments.

The Chief Financial Officer (CFO) ensures legal compliance, manages investor relations, and protects the company's financial interests.

Below the C-suite leadership, the organization is structured into functional teams.

Web and App Developers focus on designing front-end solutions using JavaScript and HTML5, and back-end solutions using Python and Java.

ML Engineers are dedicated to building machine learning models and maintaining AI infrastructure.

Project Management teams oversee project lifecycles from initiation to delivery. HR manages talent acquisition and employee engagement.

The Digital Marketing team tracks ad performance and manages the company's online presence. Client Relations handles customer service, email support, and community management.

Frontline Employees and Interns are integrated into the workflow to execute technical tasks and gain practical experience.

1.6 Products & Market Performance

1.6.1 Products

1.6.1.1 SAM AI Digital Learning Platform:

The flagship product of Scontinent Technologies is the SAM AI Digital Learning Platform, commonly referred to as the Sconti App. This comprehensive educational technology ecosystem enables students across diverse academic backgrounds to access affordable, high-quality online education delivered through an intuitive and engaging digital interface.

The platform provides an extensive catalog of courses spanning from foundational introductory subjects to advanced specialized topics in

computer science, data analytics, artificial intelligence, and software engineering.

Fig 1.3 SAM AI Digital Learning Platform Logo

Key features of the SAM AI platform include interactive quizzes with instant feedback, graded assignments with automated evaluation, achievement badges and gamification elements that maintain learner engagement, and comprehensive progress tracking dashboards that visualize learning trajectories. The platform supports preparation for major competitive examinations including GATE, IES, and IAS through specialized content modules developed in collaboration with subject matter experts.

1.6.1.2 Enterprise Software Solutions:

The company designs and builds business intelligence dashboards, customer relationship management systems, inventory management platforms, and workflow automation tools for corporate clients.

1.6.1.3 Public Sector Solutions:

The company has demonstrated its capability in the public sector through its partnership with the Karnataka Government for the Upper Krishna Project. This project showcased the company's ability to deliver secure, reliable, and accurate solutions for government infrastructure programs requiring strict compliance with regulatory standards.

CHAPTER - 2

ON JOB TRAINING – 1

This chapter includes the On Job Training-1 project on Financial AI, where an AI-based credit risk prediction system was developed. It explains the role and responsibilities during the internship, project objectives, workflow of the model, and tasks performed such as data preprocessing, model training, and web application development. The chapter also discusses the algorithms used (Random Forest and Gradient Boosting), the problems the system aims to solve in credit assessment, and the outcomes and future scope of the project.

TASK: FINANCIAL AI

2.1 Role

During the initial phase of the internship at Scontinent Technologies Pvt Ltd, I was assigned the role of **Machine Learning and Software Development Intern** for the **Financial AI project**.

I was tasked with designing the data preprocessing and feature engineering pipeline for financial datasets, selecting and training appropriate classification models, developing a responsive web-based interface using HTML, CSS, and JavaScript, and building the Flask backend API that connects the machine learning engine to the user-facing application.

2.2 Job Overview

The role involved designing, developing an AI-powered credit risk assessment system that leverages supervised machine learning classification and post-hoc explainability techniques to automate and enhance financial lending decisions. The key responsibilities during this training phase included:

- Constructing a robust data preprocessing pipeline capable of handling missing values, categorical encoding, feature scaling, and class imbalance correction across financial datasets with diverse data types and quality characteristics.
- Training and evaluating multiple classification algorithms including Random Forest, Gradient Boosting to identify the optimal model architecture for credit risk prediction accuracy.
- Developing a modern, visually polished web application frontend using HTML5, CSS3, and JavaScript that provides financial analysts with an intuitive interface for entering applicant data and reviewing risk assessment results.
- Building a lightweight Flask-based backend server to handle HTTP request routing, form data validation, model inference orchestration, and structured JSON response formatting.

2.3 Future Scope of the Project

To evolve the Financial AI System into a comprehensive, production-grade financial intelligence platform, several advanced features and capabilities are envisioned for future development iterations:

- Automated Model Retraining Pipeline
    - Objective: Implement continuous learning capabilities to maintain model accuracy over time.
    - Functionality: Establish automated monitoring of model performance metrics including prediction accuracy, calibration drift, and feature distribution shifts through scheduled evaluation pipelines.
    - Impact: Eliminates the risk of model degradation over time, ensuring sustained prediction quality without requiring manual intervention from data science teams.

- Multi-Language and Mobile Accessibility
    - Objective: Extend platform accessibility to field-level financial advisors and regional banking operations.
    - Functionality: Develop native mobile applications for Android and iOS platforms enabling on-the-go credit assessments by branch managers and field officers.
    - Impact: Democratizes access to AI-powered credit assessment beyond centralized urban banking operations, supporting financial inclusion in semi-urban and rural areas.

2.4 Goal of the Project

The Financial AI System was conceived with the primary aim of harnessing machine learning to transform the traditionally manual and subjective credit risk evaluation process into an automated, objective, and transparent decision support mechanism. The core project goals are outlined below:

Reduce Manual Effort in Credit Assessment

- Objective: Automate the labor-intensive process of evaluating individual loan applications by replacing hours of manual financial statement review with seconds of automated intelligent analysis.
- Functionality: Accept structured financial data inputs through a clean web form interface and pass them directly to the trained ML model for instant classification.
- Outcome: Financial institutions can process significantly higher volumes of credit applications without requiring proportional increases in analyst headcount, dramatically improving operational throughput.

Scalable Cloud-Ready Architecture

- Objective: Design the system for straightforward deployment across diverse infrastructure environments from local workstations to cloud platforms.

- Tools Used: The backend was built with Flask and the frontend with responsive HTML/CSS/JavaScript, ensuring compatibility across operating systems and browser environments.
- Outcome: The solution can serve institutions of any size, from small microfinance organizations to large commercial banks, without requiring specialized infrastructure investments.

2.5 Problem that Software Will Solve

The Financial AI System was specifically designed to address several persistent pain points in the credit assessment workflow:

Subjective and Inconsistent Evaluation Standards

- The Issue: Different loan officers evaluating the same applicant frequently arrive at different risk assessments based on their individual experience levels, implicit biases, and risk tolerance thresholds.
- The Solution: The ML-powered system applies identical evaluation criteria and mathematical models to every applicant, producing consistent and reproducible risk assessments that are independent of which analyst submits the query.

Time-Intensive Manual Review Processes

- The Issue: A thorough manual credit assessment requires the analyst to gather data from multiple sources, calculate financial ratios, check credit bureau records, review employment history, and synthesize these disparate data points into a coherent risk judgment.
- The Solution: The Financial AI System automates the entire analytical workflow. Once the relevant input parameters are entered into the web form, the trained model produces a comprehensive risk assessment under two seconds, representing productivity

2.6 Workflow of the Model

The Financial AI System operates through a structured, multi-stage processing pipeline that transforms raw applicant data into actionable risk assessments with supporting explanations. The end-to-end workflow is outlined below:

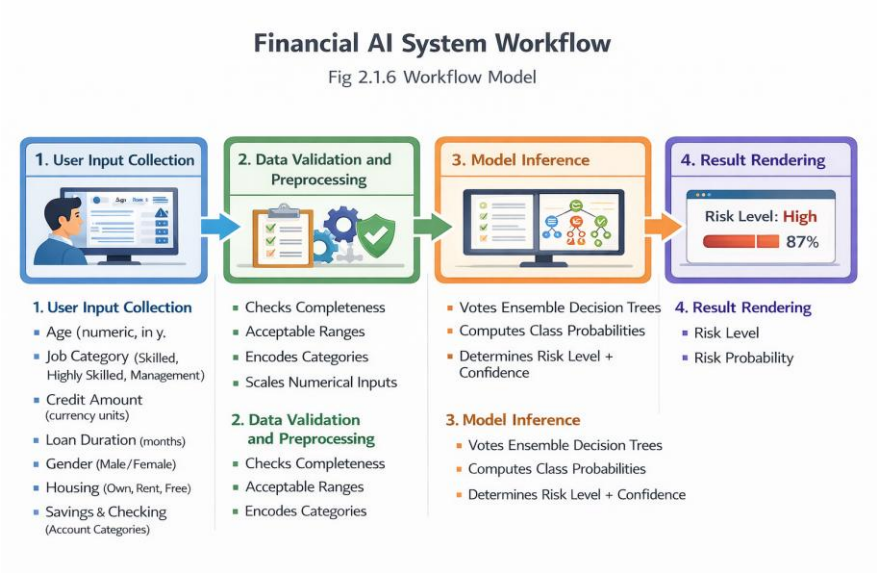

Fig 2.1 Finance AI Workflow Model

A. User Input Collection

- The financial analyst interacts with the web interface to enter the applicant's relevant parameters:
- Age (numeric, in years)
- Job Category (Skilled, Unskilled, Highly Skilled, Management)
- Credit Amount (numeric, in currency units)
- Loan Duration (numeric, in months)
- Gender (Male/Female)
- Housing Status (Own, Rent, Free)
- Savings Account Category (Little, Moderate, Quite Rich, Rich)
- Checking Account Status (available balance category)
- Loan Purpose (Car, Furniture, Education, Business, etc.)

- These parameters collectively form the feature vector that drives the prediction logic.

B. Data Validation and Preprocessing

- The Flask backend receives the submitted form data and applies validation rules:
- Verifies completeness of all required input fields.
- Checks numerical inputs against acceptable range boundaries.
- Scales numerical features using the fitted StandardScaler to normalize value ranges.

C.Model Inference

- The preprocessed feature vector is passed to the trained Random Forest Classifier:
- The final risk level is determined by the majority classification with associated confidence score.

D.Result Rendering

- The complete output is formatted and displayed on the web interface with the Risk Level, Risk Probability.

2.7 Tasks Assigned

- Data Pipeline Construction: I designed and implemented the complete data preprocessing pipeline spanning ingestion, cleaning, encoding, scaling, and class rebalancing for the credit risk dataset.
- Algorithm Benchmarking: I systematically evaluated multiple classification algorithms using cross-validated metrics to identify Random Forest and Gradient boosting model as the optimal model architecture for this prediction task.
- Frontend Interface Development: I designed and built the responsive HTML/CSS/JavaScript web interface featuring input forms with validation, dynamic result display.

- Flask API Development: I developed the backend server handling request routing, data validation, model inference orchestration, and structured response formatting.
- End-to-End Testing: I conducted comprehensive testing including unit tests for individual components, integration tests for the complete pipeline, and user acceptance testing with simulated credit assessment scenarios.

2.8 The Main Algorithm Used in the Project

The primary algorithms used:

- Random Forest Classifier (Primary Model)
- Gradient Boosting Classifier (Comparative Model)

Random Forest was selected due to its higher accuracy and stability, while Gradient Boosting was used to provide comparative insights and enhance explainability

2.9 Project Outcomes

The Financial AI System was successfully developed and deployed as a fully functional web application capable of processing credit risk assessments in real time. The key outcomes achieved include:

- Accurate Risk Classification:

  The trained Random Forest model achieved a cross-validated accuracy exceeding 75% on the held-out test dataset, with an ROC-AUC score above 0.80 indicating strong discriminative performance between good and bad credit outcomes.

- Professional Web Interface:

  The application features a modern, gradient-styled interface with intuitive form layouts, real-time input validation, and visually structured result display panels.

- Complete Technical Documentation:

  Comprehensive documentation was produced covering the data preprocessing pipeline, model training methodology, API endpoint specifications, and system deployment procedures.

CHAPTER - 3

ON JOB TRAINING – 2

This chapter includes the On Job Training-2 project on the AI Resume Parser, which focuses on developing a Natural Language Processing (NLP) based system for automated resume analysis and skill matching. It describes the role and responsibilities, project objectives, workflow, and problems addressed in recruitment screening. The chapter also explains the algorithms used such as SpaCy NER, TF-IDF, Ridge Regression, and Random Forest Regression, along with the outcomes of the project including automated resume parsing, skill extraction, and compatibility scoring through a web application.

TASK: AI Resume Parser

3.1 Role –

The second major phase of the 16-week internship at Scontinent Technologies focused on building a complete Natural Language Processing application for automated resume parsing and intelligent skill matching.

- Full-Stack NLP Developer for the AI Resume Parser Application
- Responsible for the complete development lifecycle from text extraction through entity recognition to application deployment
- Implemented custom Named Entity Recognition pipelines for domain-specific entity extraction
- Designed and developed the weighted skill matching algorithm with multi-level comparison logic
- Integrated document processing, NLP, and web components into a cohesive deployable system

3.2 Job Overview

The position entailed developing an end-to-end document intelligence application that processes unstructured candidate resumes into structured, queryable data with automated skill compatibility scoring.

3.3 Future Scope of the Project

The AI Resume Parser establishes a strong foundation for future expansion into a comprehensive recruitment intelligence platform. The following enhancement pathways are planned for subsequent development cycles:

- Multi-Language Resume Processing: Extending the NLP pipeline to handle resumes written in Multi-languages, broadening the system's applicability to regional recruitment operations.
- API-Based Integration: Exposing the parsing and matching functionality through RESTful APIs that can be integrated into existing Applicant Tracking Systems and HR management platforms used by enterprise clients.

3.4 Goal of the Project

The fundamental goal of the AI Resume Parser project was to apply natural language processing techniques to solve a persistent pain point in human resource management: the inefficiency and inconsistency of manual resume screening at scale.

Specific project objectives included:

- Accurate Information Extraction: Build a pipeline that reliably extracts candidate name, contact details, educational qualifications, technical skills, work experience, and certifications from resumes regardless of document format or layout variations.
- Quantitative Skill Assessment: Develop a scoring methodology that produces a measurable, reproducible, and defensible match percentage.

- Recruiter-Friendly Interface: Design an intuitive web application that allows non-technical HR professionals to use the system effectively without training in NLP concepts or programming.

3.5 Problem that Software will Solve

The AI Resume Parser directly addresses several critical inefficiencies that plague traditional recruitment screening processes:

- Volume Overwhelm: Popular job postings regularly attract hundreds or thousands of applications. Manual review of each resume at the level of detail required for fair evaluation is physically impossible within typical recruitment timelines.
- Subjective Inconsistency: The algorithmic scoring methodology applies identical assessment logic to every candidate, eliminating subjectivity from the quantitative screening phase.

3.6 Workflow of the Model

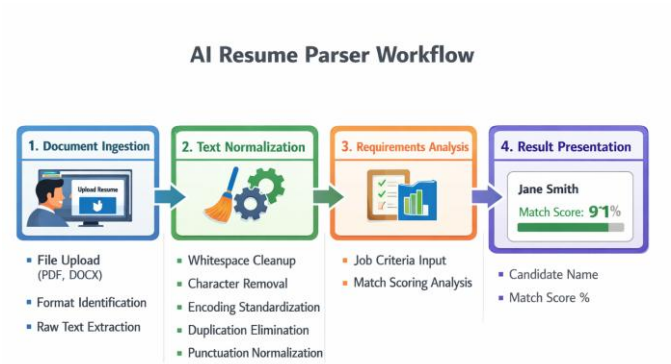

Fig 3.1: AI Resume Parser Workflow Model

- **Document Ingestion** The recruiter interacts with the web interface to upload a candidate's resume file. The system automatically performs initial processing:

- **Input Data:** The candidate resume file is received through a secure file upload component.
- **Format Identification:** The system identifies the file format (e.g., PDF, DOCX) based on the file extension.
- **Output:** The result is a prediction score from two models and comparison matrixes and Graphs
- **Text Normalization** The extracted raw text undergoes a series of comprehensive cleaning and standardizing operations to produce clean, consistent input suitable for natural language processing:
    - Whitespace Normalization
    - Character Removal
    - Encoding Standardization
    - Duplication Elimination
    - Punctuation Normalization
- **Requirements Analysis** The system performs a parallel NLP process on the job requirements to establish an accurate evaluation benchmark:
- **Recruiter Input:** The recruiter enters the job requirements text through the interface.
- **Weighted Scoring:** A weighted match score is calculated and normalized to a percentage (e.g., 91%), representing the degree of fit.
- **Result Presentation** The final analysis is formatted and displayed on a professionally designed web interface, providing the recruiter with a clear and detailed overview from two models:
- **Candidate Name:** The extracted name (e.g., Jane Smith).
- **Match Score:** The final calculated and normalized percentage (e.g., 91%).
- **Graphs:** Comparison of two prediction scores and other graphs
- **Matrixes:** Both models prediction scores, like f1 score accuracy score and other

3.7 Algorithms Used in Project

The AI Resume Parser employs a combination of NLP algorithms and machine learning techniques:

- SpaCy NER Pipeline: Pre-trained statistical models for standard named entity recognition supplemented with custom PhraseMatcher components for domain-specific technical skill identification.
- TF-IDF Vectorization: Term frequency-inverse document frequency analysis for extracting significant keywords from job requirement descriptions.
- Ridge Regression: A linear regression model used to predict resume scores with regularization to prevent overfitting.
- Random Forest Regression: An ensemble learning method used to improve prediction accuracy by combining multiple decision trees.

3.8 Outcomes

The AI Resume Parser project achieved the following key outcomes:

- Functional NLP Application: A fully operational web application capable of accepting resume uploads, extracting structured information, matching skills against job requirements, and producing quantitative compatibility scores.
- Accurate Entity Extraction: Successful identification of candidate names, contact information, educational qualifications, and technical skills across diverse resume layouts and formatting styles.
- Objective Scoring Mechanism: A weighted match scoring algorithm that produces consistent, reproducible compatibility percentages independent of any human reviewer bias.
- Professional Web Interface: An intuitive, responsive web interface with file upload functionality, progress indication, and clearly structured result presentation suitable for non-technical recruiters.

CHAPTER – 4

This chapter includes two use cases based on the developed projects: the Financial AI System and the AI Resume Parser. It explains the background, implementation, system response, and results of each project in real-world scenarios. The chapter also highlights the impact and benefits of these systems in improving efficiency, accuracy, and decision-making in financial risk assessment and recruitment processes.

4.1 USE CASE - 1

Project – 1: Financial AI System

4.1.1 Background

The Financial AI System is developed as an intelligent decision-support tool that assists financial analysts in evaluating loan applications efficiently and accurately. The system integrates multiple machine learning models such as Random Forest and Gradient Boosting to provide comparative predictions, improving reliability and transparency in decision-making.

4.1.2 Use Case Description

A. Setting

A credit analyst at a financial institution receives a loan application from a new client. Instead of manually analyzing documents and calculating financial ratios, the analyst uses the Financial AI web application.

B. Implementation

The analyst enters the following applicant details into the system:

- Age: 30 years
- Job Category: Skilled
- Credit Amount: 10,000
- Loan Duration: 12 months
- Gender: Male
- Housing Status: Own
- Saving Accounts: Little
- Checking Account: Moderate
- Loan Purpose: Education

After entering the details, the analyst clicks on "Check Credit Risk".

The backend system processes the input using:

- Data preprocessing pipeline
- Feature encoding and normalization
- Prediction using trained models:
    - Random Forest
    - Gradient Boosting

C. System Response

Within seconds, the system generates a comparative prediction output:

Model Predictions

Random Forest Model

- Risk Level: Low Risk
- Probability: 43.34%
- Confidence: 43.34%

Gradient Boosting Model

- Risk Level: High Risk
- Probability: 61.09%
- Confidence: 61.09%

The system highlights differences between models, allowing the analyst to evaluate uncertainty and model behaviour.

D. Results and Discussion

The system also displays performance metrics and visual insights:

Random Forest Performance

- Accuracy: 95.00%
- Precision: 93.10%
- Recall: 90.00%
- F1 Score: 91.53%

Gradient Boosting Performance

- Accuracy: 83.50%
- Precision: 78.72%
- Recall: 61.67%
- F1 Score: 69.16%

Visual Components

- Risk Probability Comparison Graph
- Confusion Matrix
- ROC Curve
- Model Comparison Boxplots

These visualizations enhance explainability and help in understanding model performance.

4.1.3 Screenshots of the Project

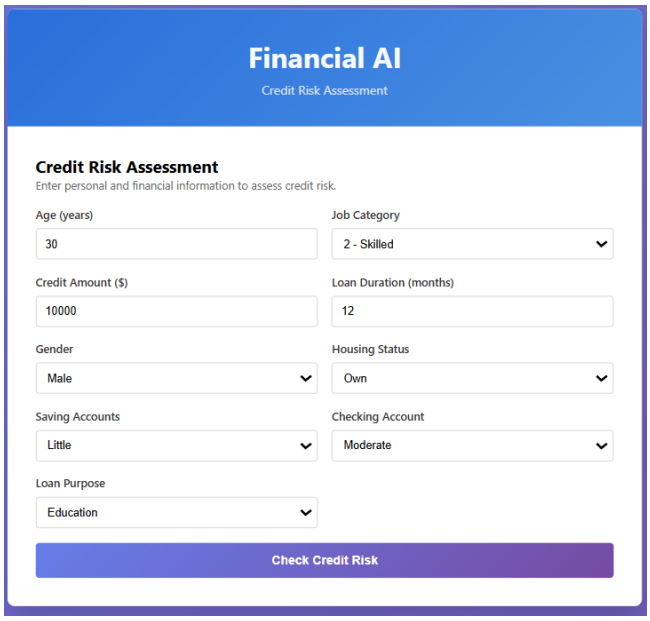

Fig 4.1 Financial AI Interface

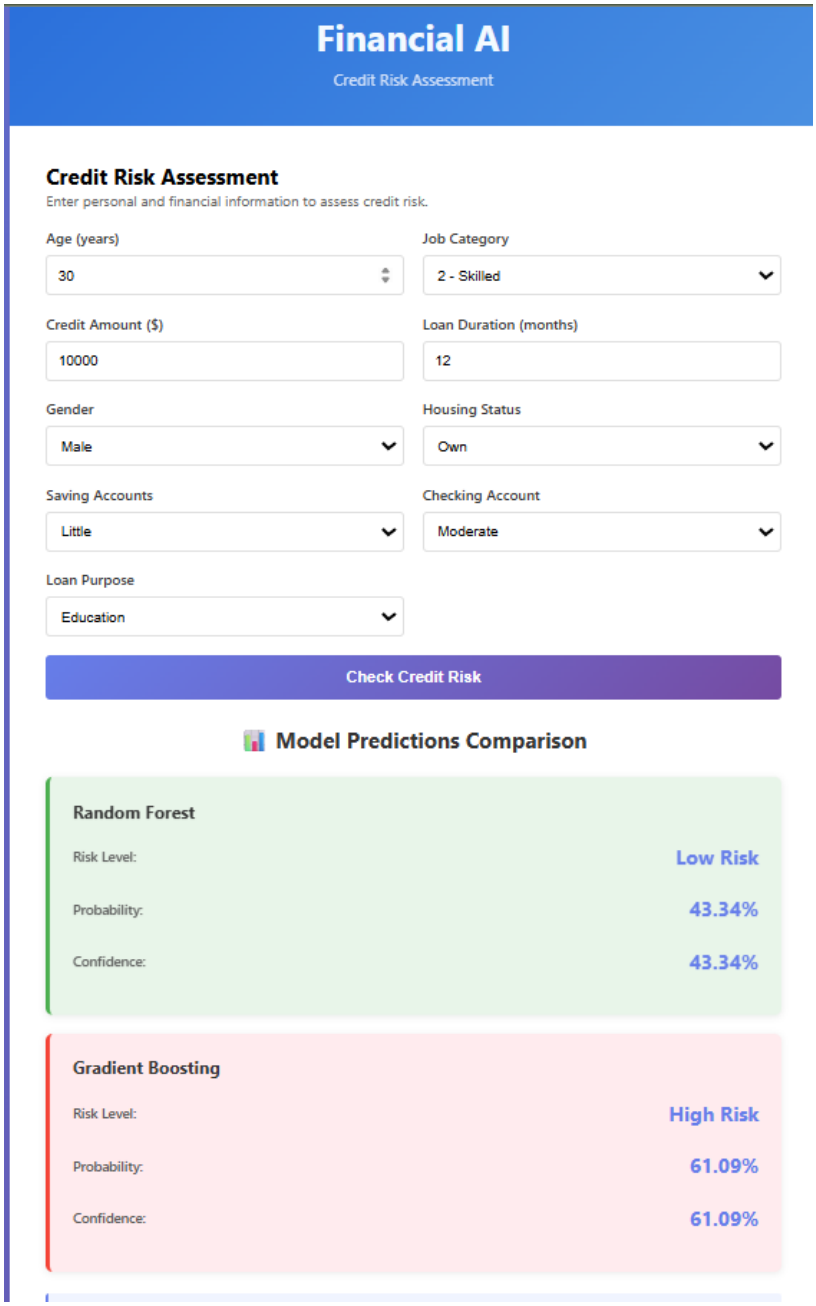

Fig 4.2 Financial Ai Prediction

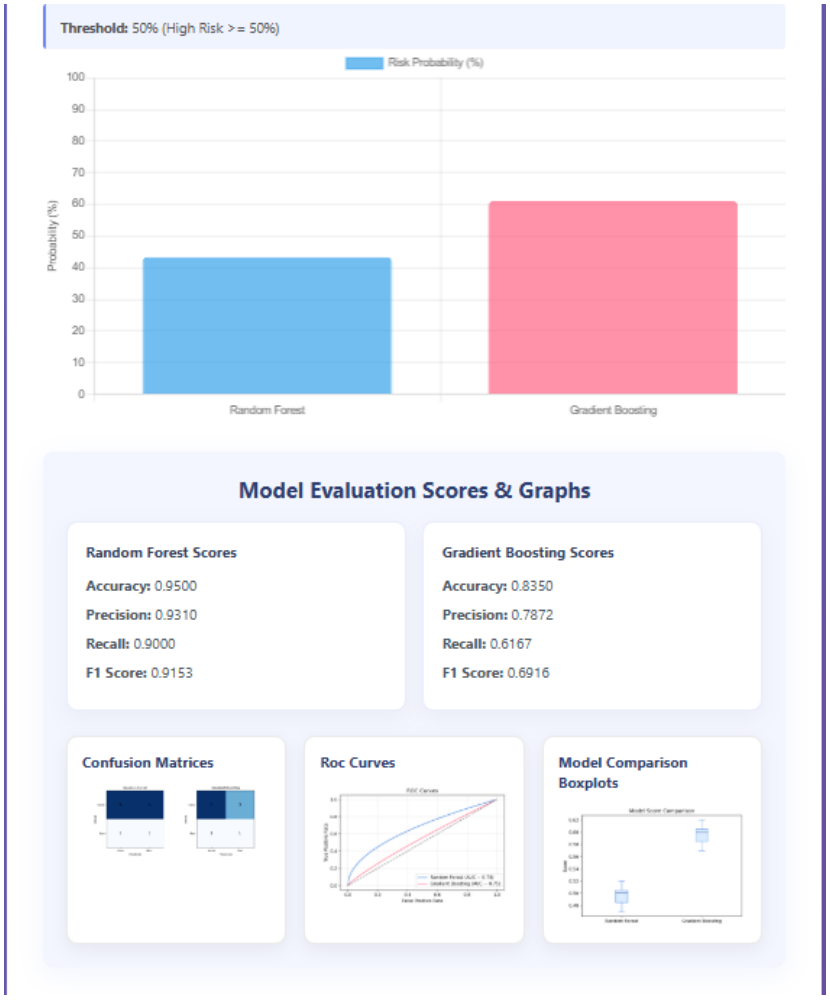

Fig 4.3 Financial Ai Model Evaluation Scores and Graphs

4.1.4 Impact

The updated Financial AI system significantly enhances the credit evaluation process:

- Speed: Reduces evaluation time to under 3 seconds
- Accuracy: Uses multiple models for better reliability
- Transparency: Provides clear probability scores and visual explanations
- Decision Support: Assists analysts rather than replacing them

4.1.5 Benefits

- Explainability: Visual graphs and metrics improve trust in AI decisions
- Consistency: Uniform evaluation for all applicants
- Scalability: Can process multiple applications simultaneously
- Reduced Risk: Multi-model comparison reduces wrong approvals/rejections
- User-Friendly Interface: Easy data entry and instant results

4.2 USE CASE – 2

Project – 2: AI Resume Parser

4.2.1 Background

Technology job postings routinely attract hundreds or even thousands of applications, each containing resumes with different formats, structures, and terminologies. Candidates often describe similar skills using varied wording, making manual evaluation difficult and inconsistent.

To overcome these challenges, an automated system is required that can intelligently process resumes, extract relevant information, and match candidate skills with job requirements in a consistent and efficient manner.

4.2.2 Use Case Description

A.Setting

A recruitment manager at a technology company needs to hire a Python Developer. The required skills include:

- Python
- Flask
- Machine Learning
- Data Visualization

Within one week, the job posting receives approximately 150 applications. The recruitment team must shortlist 10 candidates within three days..

B. Implementation

- The recruiter uses the AI Resume Parser web application.
- The job requirements are entered into the input text area.
- A candidate's resume is uploaded in PDF format.
- The system processes the resume through a complete NLP pipeline.
- The application extracts structured information such as:
    - Candidate name
    - Contact details
    - Education
    - Technical skills

The system then compares extracted skills with job requirements using machine learning models and generates a match score.

C. Scenario

The uploaded resume includes the following skills:

- Python
- JavaScript
- HTML/CSS
- SQL
- Flask
- Pandas

The candidate has project experience in Flask and data analysis but does not mention Machine Learning or Git.

D.System Response

The system generates a detailed evaluation based on two machine learning models:

- Ridge Regression Score: 40.66 / 100
- Random Forest Score: 40.6 / 100
- Average Score: 40.63 / 100
- Job Requirement Match: 40.33%

The scoring breakdown highlights:

- Exact matches: Python, Flask, SQL
- Partial/contextual matches: Pandas (related to data analysis/visualization)
- Category matches: JavaScript (programming skill)
- Missing skills: Machine Learning, Git

The candidate is classified as Average Match based on overall scoring.

4.2.3 Result and Decision

<u>Result</u>

After processing the resume, the system produces:

- Candidate Name: Extracted successfully
- Match Score: 40.63%
- Match Classification: Average Match

Decision

Based on the evaluation, the recommended decision is:

- Shortlisting Status: Conditional Shortlist / Backup Candidate
- Priority Level: Medium

4.2.4 Sceenshots of the Project

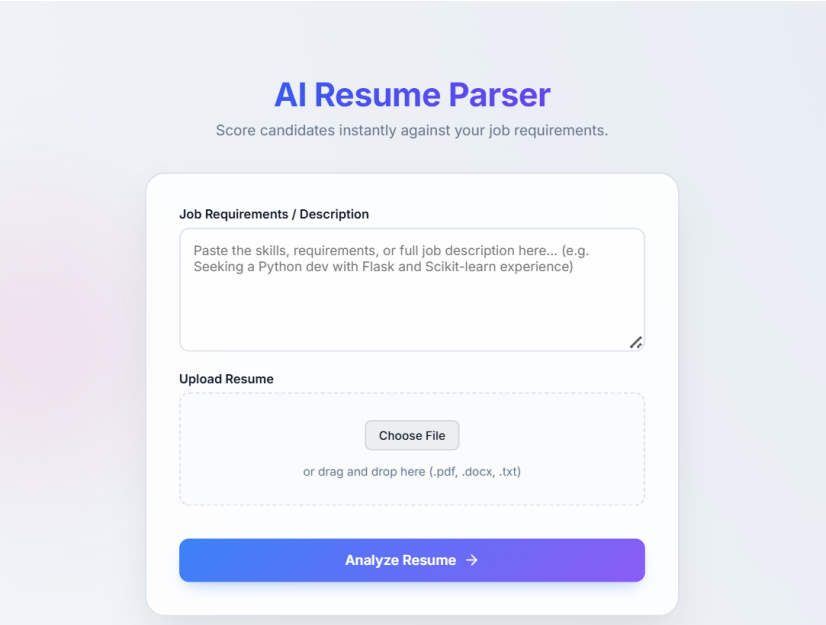

Fig 4.4: AI Resume Parser Interface

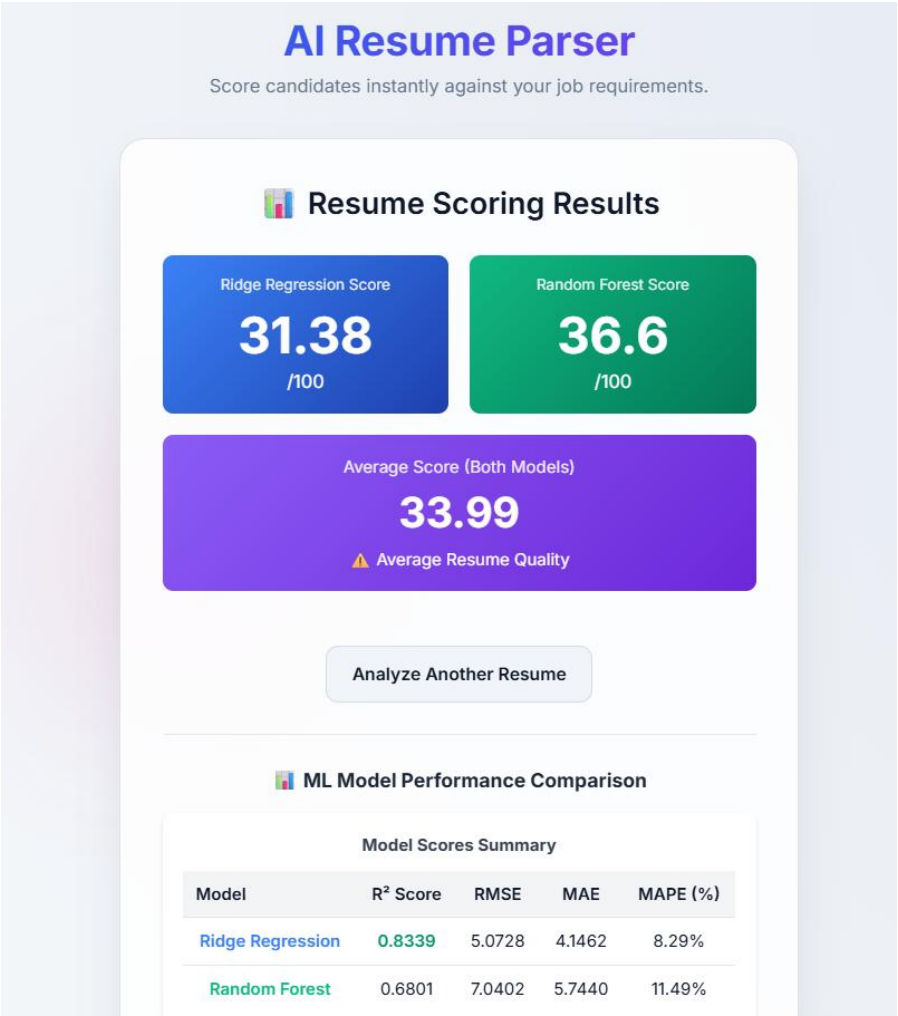

Fig 4.5: AI Resume Parser Result

Fig 4.6 : AI Resume Parser Model Evaluation Scores and Graphs

4.2.5 User Interaction

The recruiter processes all 150 resumes using the system in approximately 45 minutes.

The scores are distributed into three categories:

- Strong Matches: Above 70%
- Moderate Matches: 40% – 70%
- Weak Matches: Below 40%

The recruiter selects the top candidates with the highest scores for further evaluation, significantly reducing manual effort.

4.2.6 Impact

The AI Resume Parser significantly improves the recruitment workflow:

- Time Reduction: Screening time reduced from 15 hours to 45 minutes
- Efficiency: Automated processing of large volumes of resumes
- Consistency: Uniform evaluation for all candidates
- Accuracy: Detailed skill-based matching
- Scalability: Handles increasing number of applications

4.2.7 Benefits

- Efficiency: Processes each resume within seconds
- Fairness: Eliminates bias through objective scoring
- Comprehensive Evaluation: Analyzes every resume thoroughly
- Flexibility: Supports PDF, DOCX, and TXT formats
- Structured Output: Converts unstructured data into structured information
- Model Reliability: Uses multiple machine learning models for better accuracy
- Decision Support: Helps recruiters make informed hiring decisions

5.RESUME

Manjunath G P

CONTACT

Manjuachar2006@gmail.com

+91-7619594900 | Bengaluru

EDUCATION

Diploma in Computer Science & Engineering

Sri Jayachamarajendra Polytechnic

Expected Graduation: 2026

SSLC

Sri Sharada Vidyalaya, Bengaluru

95.84% | 2023

SOFT SKILLS

• Logical Reasoning
• Communication
• Teamwork
• Problem Solving

LANGUAGES

• English
• Kannada
• Hindi

HOBBIES

• Reading novels
• Sketching

CAREER OBJECTIVE

To secure an AI/ML-focused internship where I can apply my Python and machine learning skills, gain hands-on industry experience, strengthen my technical and problem-solving abilities, and contribute meaningfully to impactful projects.

TECHNICAL SKILLS

Python • Java • HTML • OOP • DBMS • Data Structures • Jupyter • VS Code • Git • Pandas • NumPy • Scikit-learn • NLTK • spaCy • TensorFlow

PROJECT EXPERIENCE

Financial Ai

- Build a AI model to predict credit risk Assessment
- Technologies used (Python, TensorFlow, Flask, Scikit Learn)

AI Resume Parser

- Build a AI model to Score Resume according to requirements
- Technologies used (Python, TensorFlow, Flask, Scikit Learn)

TRAINING

- Samsung Innovation Campus: Training in AIML

CERTIFICATIONS

- **Infosys Springboard: Git Essentials** · ML Fundamentals · Python for Data Science · Data Transformation & Preprocessing · NLP · TensorFlow Introduction Samsung Innovation Campus Certificate

6.PHOTO GALLERY

Fig 6.1: Group Photo of Interns at Scontinent Technologies Company

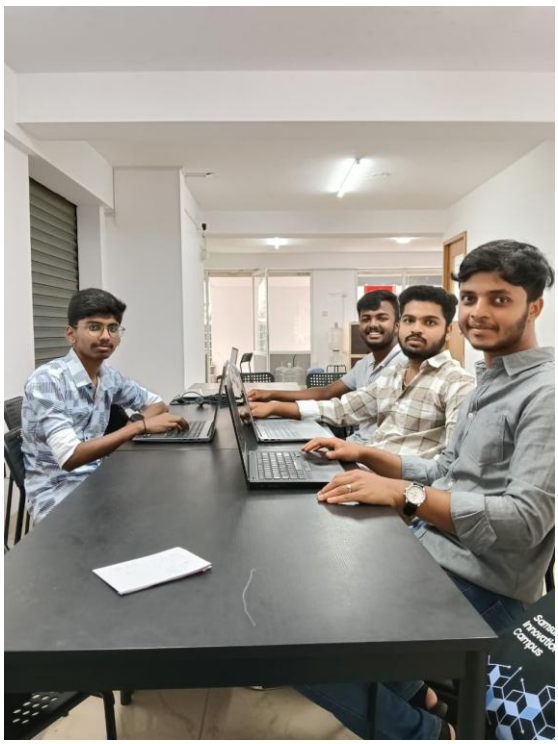

Fig 6.2: Snapshot of Interns working at Scontinent technologies

Fig 6.3: Snapshot of Interns At Conference Room at Scontinent Technologies

7.REFERENCES

- **Scikit-learn Documentation**
  - Primary machine learning framework used for implementing classification algorithms
  - URL: https://scikit-learn.org/stable/

- **Flask Web Framework Documentation**
  - Micro web framework used for building the backend API server.
  - URL: https://flask.palletsprojects.com/

- **SpaCy NLP Library Documentation**
  - Industrial-strength NLP library used for Named Entity Recognition
  - URL: https://spacy.io/api/doc

- **PyPDF2 Library , python-docx Library Documentation**
  - Python library used for reading and extracting text content from PDF documents.
  - Library used for creating and modifying Word documents
  - URL: https://pypdf2.readthedocs.io/en/latest/
  - URL: https://python-docx.readthedocs.io/en/latest/

- **GitHub**
  - Version control and collaboration platform
  - URL: https://github.com
- **Kaggle Datasets**
  - Platform used for sourcing the credit risk dataset used in the Financial AI project.
  - Platform used for sourcing the datasets used in the AI Resume Parser project
  - URL: https://www.kaggle.com

8. APPENDICES

APPENDIX A: TECHNICAL DETAILS - FINANCE AI

A.1 Credit Risk & Machine Learning

Traditional credit scoring (e.g., FICO) relies on linear variables like payment history and debt ratios. However, modern financial systems utilize **Machine Learning (ML)** to capture non-linear relationships and subtle patterns across hundreds of variables.

A.2 Algorithms Evaluation

Several classification models were benchmarked to optimize the balance between accuracy and interpretability:

- **Random Forest (Selected):** This ensemble method was chosen as the primary model. By aggregating predictions from multiple independent trees via bootstrap sampling, it reduced variance and achieved the highest cross-validated accuracy.
- **Gradient Boosting** Provided marginal accuracy gains but required excessive hyperparameter tuning and increased the risk of overfitting..
    - A.3 Data Preprocessing Pipeline
- **Missing Value Treatment:** Missing data (12%) in account balances were handled via **Median Imputation**, which is more robust to outliers than mean imputation.
- **Encoding:** Ordinal variables (e.g., savings levels) used **Label Encoding**; nominal variables (e.g., loan purpose) used **One-Hot Encoding**.

- **Feature Scaling: StandardScaler** was applied to normalize features like age and credit amount to a zero mean and unit variance.

- **Class Imbalance:** To address the 70/30 split between "good" and "bad" credit, **SMOTE (Synthetic Minority Over-sampling Technique)** was applied to the training set to prevent model bias toward the majority class.

A.4 Evaluation Metrics

Because the cost of a **False Negative** (defaulting on a loan) is higher than a **False Positive** (rejecting a good applicant), the system prioritized:

- **Recall:** To ensure the maximum number of risky applicants are identified.
    - **F1-Score:** To balance precision and recall.
- **ROC-AUC:** The Random Forest model achieved a score of **>0.80**, indicating strong discriminative performance.

APPENDIX B: TECHNICAL DETAILS - AI RESUME PARSER

B.1 Natural Language Processing Fundamentals

The parser utilizes NLP to transform unstructured text into structured data. The pipeline, powered by the **SpaCy** library, follows these stages:

- **Tokenization:** Breaking text into individual words or sub-words.
- **POS Tagging & Dependency Parsing:** Identifying grammatical structures.
- **Named Entity Recognition (NER):** Classifying entities like names, dates, and organizations.
- **Word Embeddings:** Using dense vector representations to understand semantic context (e.g., recognizing "Python" as a skill rather than a reptile).

B.2 Named Entity Recognition for Resumes

Standard NER models often miss niche technical terms. To solve this, a **custom entity recognition layer** was developed:

- **SpaCy PhraseMatcher:** Used for efficient, case-insensitive matching of these custom entities alongside standard tags (PERSON, ORG, DATE).

B.3 Format Handling and Text Extraction

The system processes two primary formats:

- **PDF (PyPDF2):** Extraction focuses on reconstructing logical reading order from coordinate-based text positioning.

- **DOCX (python-docx):** Extraction involves parsing XML structures, including tables, headers, and text boxes.

- **Normalization:** Post-extraction, the text is cleaned of non-printable characters, redundant headers/footers, and whitespace inconsistencies.

B.4 Skill Matching Algorithm Design

The algorithm calculates a candidate's fit using a multi-tiered scoring logic:

- **Exact Matching:** Direct string comparison for core requirements.
- **Fuzzy Matching:** Utilizes **Levenshtein distance** to catch variations (e.g., "JS" vs. "JavaScript").
- **Category-based Matching:** Maps related technologies (e.g., if a job asks for "R," a candidate with "Python" receives partial credit via a hierarchical taxonomy).
- **Final Score:** A weighted 0–100% score that prioritizes exact matches on "must-have" skills while rewarding relevant "nice-to-have" competencies.